

NT-C-028 | Cloud Computing | Course Knowledge Profile

Identity, purpose, prerequisites, scope and boundaries

Category	Cloud Foundations
Target learners	Learners needing vendor-neutral cloud concepts before platform specialisation.
Prerequisites	Basic computer, networking and operating-system awareness.
Approved duration	10-14 weeks / 100-150 guided hours.
Final outcome	Design and operate a secure, resilient cloud solution using core compute, storage, network and automation concepts.
Assessment profile	Infrastructure/Security

Approved tools / platforms

Linux, virtualization/container tools, cloud-provider sandbox, networking tools and Git.

Knowledge boundaries

Include	Quality rule	Exclude / escalate
The eight approved modules, four sections, named tools, projects and mapped entry-level roles.	Require least privilege, authorised labs, configuration evidence, logs, monitoring, rollback and documented troubleshooting.	Never recommend unsafe production testing or treat a scanner output as a verified finding without validation.
Current product/platform procedures only when version and date are known.	Use primary documentation and record the source version.	Do not invent current commands, prices, tax rates, policies or certification status.
Evidence-based career guidance.	Cite tests, projects, capstone, viva and verified resume evidence.	No guaranteed job, placement, salary, interview or employer claim.

LLM answer pattern: identify the course and version, answer from approved records, cite record IDs, distinguish verified facts from guidance, and state any missing evidence.

NT-C-028 | Cloud Computing | NT-C-028-S01 Deep Knowledge

Modules 1-2 | objectives, concepts, evidence and remediation

Section competency	IaaS, PaaS, SaaS, deployment models and shared responsibility; VMs, images, containers and orchestration basics
Completion evidence	Concept map; Virtual lab
Section weight	20% of the weighted mock-test average
Assessment	10 MCQ (30 marks) + 4 troubleshooting scenarios (30) + 1 authorised practical lab (40)

Module 1 - Cloud concepts

Objective: IaaS, PaaS, SaaS, deployment models and shared responsibility

Observable evidence: Concept map

Concept	Approved knowledge statement	Application behaviour	Evidence
IaaS	A defined competency within Cloud Computing. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when IaaS is appropriate; apply it in a Cloud Computing task; verify the result.	Concept map
PaaS	A defined competency within Cloud Computing. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when PaaS is appropriate; apply it in a Cloud Computing task; verify the result.	Concept map
SaaS	A defined competency within Cloud Computing. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when SaaS is appropriate; apply it in a Cloud Computing task; verify the result.	Concept map
deployment models	A defined competency within Cloud Computing. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when deployment models is appropriate; apply it in a Cloud Computing task; verify the result.	Concept map
shared responsibility	A defined competency within Cloud Computing. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when shared responsibility is appropriate; apply it in a Cloud Computing task; verify the result.	Concept map

Module 2 - Virtualisation

Objective: VMs, images, containers and orchestration basics

Observable evidence: Virtual lab

Concept	Approved knowledge statement	Application behaviour	Evidence
VMs	A defined competency within Cloud Computing. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when VMs is appropriate; apply it in a Cloud Computing task; verify the result.	Virtual lab
images	A defined competency within Cloud Computing. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when images is appropriate; apply it in a Cloud Computing task; verify the result.	Virtual lab
containers	A packaged application process with its dependencies and an isolated runtime. In Cloud Computing, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when containers is appropriate; apply it in a Cloud Computing task; verify the result.	Virtual lab
orchestration basics	A defined competency within Cloud Computing. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when orchestration basics is appropriate; apply it in a Cloud Computing task; verify the result.	Virtual lab

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Concept map; Virtual lab, return to Modules 1-2, complete one guided practice and then use a new equivalent test form.

NT-C-028 | Cloud Computing | NT-C-028-S02 Deep Knowledge

Modules 3-4 | objectives, concepts, evidence and remediation

Section competency	Addressing, DNS, routing, firewalls and load balancing; Object, block, file and managed databases
Completion evidence	Network diagram; Service comparison
Section weight	25% of the weighted mock-test average
Assessment	10 MCQ (30 marks) + 4 troubleshooting scenarios (30) + 1 authorised practical lab (40)

Module 3 - Networking

Objective: Addressing, DNS, routing, firewalls and load balancing

Observable evidence: Network diagram

Concept	Approved knowledge statement	Application behaviour	Evidence
Addressing	A defined competency within Cloud Computing. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when addressing is appropriate; apply it in a Cloud Computing task; verify the result.	Network diagram
DNS	A defined competency within Cloud Computing. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when dns is appropriate; apply it in a Cloud Computing task; verify the result.	Network diagram
routing	Selection and forwarding of network paths between destinations. In Cloud Computing, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when routing is appropriate; apply it in a Cloud Computing task; verify the result.	Network diagram
firewalls	A control that allows or blocks traffic according to defined rules. In Cloud Computing, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when firewalls is appropriate; apply it in a Cloud Computing task; verify the result.	Network diagram
load balancing	A defined competency within Cloud Computing. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when load balancing is appropriate; apply it in a Cloud Computing task; verify the result.	Network diagram

Module 4 - Storage and data

Objective: Object, block, file and managed databases

Observable evidence: Service comparison

Concept	Approved knowledge statement	Application behaviour	Evidence
Object	A defined competency within Cloud Computing. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when object is appropriate; apply it in a Cloud Computing task; verify the result.	Service comparison
block	A defined competency within Cloud Computing. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when block is appropriate; apply it in a Cloud Computing task; verify the result.	Service comparison
file	A defined competency within Cloud Computing. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when file is appropriate; apply it in a Cloud Computing task; verify the result.	Service comparison
managed databases	A defined competency within Cloud Computing. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when managed databases is appropriate; apply it in a Cloud Computing task; verify the result.	Service comparison

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Network diagram; Service comparison, return to Modules 3-4, complete one guided practice and then use a new equivalent test form.

NT-C-028 | Cloud Computing | NT-C-028-S03 Deep Knowledge

Modules 5-6 | objectives, concepts, evidence and remediation

Section competency	IAM, encryption, secrets and controls; Scaling, availability, backup and disaster recovery
Completion evidence	Security checklist; Resilience plan
Section weight	25% of the weighted mock-test average
Assessment	10 MCQ (30 marks) + 4 troubleshooting scenarios (30) + 1 authorised practical lab (40)

Module 5 - Identity and security

Objective: IAM, encryption, secrets and controls

Observable evidence: Security checklist

Concept	Approved knowledge statement	Application behaviour	Evidence
IAM	Identity and access management based on users, roles, policies and least privilege. In Cloud Computing, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when iam is appropriate; apply it in a Cloud Computing task; verify the result.	Security checklist
encryption	A defined competency within Cloud Computing. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when encryption is appropriate; apply it in a Cloud Computing task; verify the result.	Security checklist
secrets	A defined competency within Cloud Computing. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when secrets is appropriate; apply it in a Cloud Computing task; verify the result.	Security checklist
controls	A defined competency within Cloud Computing. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when controls is appropriate; apply it in a Cloud Computing task; verify the result.	Security checklist

Module 6 - Reliability

Objective: Scaling, availability, backup and disaster recovery

Observable evidence: Resilience plan

Concept	Approved knowledge statement	Application behaviour	Evidence
Scaling	A defined competency within Cloud Computing. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when scaling is appropriate; apply it in a Cloud Computing task; verify the result.	Resilience plan
availability	A defined competency within Cloud Computing. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when availability is appropriate; apply it in a Cloud Computing task; verify the result.	Resilience plan
backup	A defined competency within Cloud Computing. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when backup is appropriate; apply it in a Cloud Computing task; verify the result.	Resilience plan
disaster recovery	A defined competency within Cloud Computing. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when disaster recovery is appropriate; apply it in a Cloud Computing task; verify the result.	Resilience plan

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Security checklist; Resilience plan, return to Modules 5-6, complete one guided practice and then use a new equivalent test form.

NT-C-028 | Cloud Computing | NT-C-028-S04 Deep Knowledge

Modules 7-8 | objectives, concepts, evidence and remediation

Section competency	Monitoring, logging, automation and cost management; Vendor comparison and deployable architecture
Completion evidence	Operations dashboard; Cloud solution design
Section weight	30% of the weighted mock-test average
Assessment	10 MCQ (30 marks) + 4 troubleshooting scenarios (30) + 1 authorised practical lab (40)

Module 7 - Operations

Objective: Monitoring, logging, automation and cost management

Observable evidence: Operations dashboard

Concept	Approved knowledge statement	Application behaviour	Evidence
Monitoring	Collection and interpretation of metrics, logs and events to understand system health. In Cloud Computing, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when monitoring is appropriate; apply it in a Cloud Computing task; verify the result.	Operations dashboard
logging	A defined competency within Cloud Computing. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when logging is appropriate; apply it in a Cloud Computing task; verify the result.	Operations dashboard
automation	A defined competency within Cloud Computing. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when automation is appropriate; apply it in a Cloud Computing task; verify the result.	Operations dashboard
cost management	A defined competency within Cloud Computing. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when cost management is appropriate; apply it in a Cloud Computing task; verify the result.	Operations dashboard

Module 8 - Capstone

Objective: Vendor comparison and deployable architecture

Observable evidence: Cloud solution design

Concept	Approved knowledge statement	Application behaviour	Evidence
Vendor comparison	A defined competency within Cloud Computing. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when vendor comparison is appropriate; apply it in a Cloud Computing task; verify the result.	Cloud solution design
deployable architecture	A defined competency within Cloud Computing. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when deployable architecture is appropriate; apply it in a Cloud Computing task; verify the result.	Cloud solution design

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Operations dashboard; Cloud solution design, return to Modules 7-8, complete one guided practice and then use a new equivalent test form.

NT-C-028 | Cloud Computing | Skill Ontology & Dependency Map

Atomic skills, learning order and evidence strength

Skill ID	Competency	Depends on	Evidence	Type
NT-C-028-SK01	Cloud concepts	Prerequisite foundation	Concept map	Critical
NT-C-028-SK02	Virtualisation	M01 plus earlier foundations	Virtual lab	Critical
NT-C-028-SK03	Networking	M02 plus earlier foundations	Network diagram	Supporting
NT-C-028-SK04	Storage and data	M03 plus earlier foundations	Service comparison	Supporting
NT-C-028-SK05	Identity and security	M04 plus earlier foundations	Security checklist	Supporting
NT-C-028-SK06	Reliability	M05 plus earlier foundations	Resilience plan	Critical
NT-C-028-SK07	Operations	M06 plus earlier foundations	Operations dashboard	Supporting
NT-C-028-SK08	Capstone	M07 plus earlier foundations	Cloud solution design	Critical

Evidence-strength hierarchy

Strength	Evidence source	Use
High	Trainer-observed practical, viva, working source/project files and reproducible output.	May support verified mastery when rubric threshold is met.
Medium	Scored mock tests, reviewed assignments and documented portfolio artifacts.	Supports mastery with corroboration.
Low	Resume claim, self-report, attendance or video completion without demonstration.	May trigger verification; cannot alone support Apply Now.

CareerTours must not treat attendance, time spent or content opened as skill mastery.

NT-C-028 | Cloud Computing | Assessment Knowledge & Sample Items

Non-live examples for LLM training and evaluation

Sample ID	Type	Question	Approved answer / rubric anchor
NT-C-028-S01-EX01	Evidence/application	What evidence best demonstrates the combined competency of Cloud concepts and Virtualisation?	The learner should provide Concept map; Virtual lab and explain how the work applies IaaS, PaaS, SaaS, deployment models and shared responsibility; VMs, images, containers and orchestration basics. Recall alone is insufficient.
NT-C-028-S01-EX02	Scenario/remediation	A learner reports completing Virtualisation but cannot produce Virtual lab. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-028-S01, request Virtual lab, and allow a new equivalent test only after remediation.
NT-C-028-S02-EX01	Evidence/application	What evidence best demonstrates the combined competency of Networking and Storage and data?	The learner should provide Network diagram; Service comparison and explain how the work applies Addressing, DNS, routing, firewalls and load balancing; Object, block, file and managed databases. Recall alone is insufficient.
NT-C-028-S02-EX02	Scenario/remediation	A learner reports completing Storage and data but cannot produce Service comparison. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-028-S02, request Service comparison, and allow a new equivalent test only after remediation.
NT-C-028-S03-EX01	Evidence/application	What evidence best demonstrates the combined competency of Identity and security and Reliability?	The learner should provide Security checklist; Resilience plan and explain how the work applies IAM, encryption, secrets and controls; Scaling, availability, backup and disaster recovery. Recall alone is insufficient.
NT-C-028-S03-EX02	Scenario/remediation	A learner reports completing Reliability but cannot produce Resilience plan. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-028-S03, request Resilience plan, and allow a new equivalent test only after remediation.
NT-C-028-S04-EX01	Evidence/application	What evidence best demonstrates the combined competency of Operations and Capstone?	The learner should provide Operations dashboard; Cloud solution design and explain how the work applies Monitoring, logging, automation and cost management; Vendor comparison and deployable architecture. Recall alone is insufficient.
NT-C-028-S04-EX02	Scenario/remediation	A learner reports completing Capstone but cannot produce Cloud solution design. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-028-S04, request Cloud solution design, and allow a new equivalent test only after remediation.

Live bank rule: Generate at least three times the live quantity for each section using the approved Infrastructure/Security blueprint. Human-review every scored item before initial production release.

NT-C-028 | Cloud Computing | Labs, Projects & Scoring Rubrics

Practical proof of learning and ownership

Level	Approved project	Skills evidenced	Required deliverables
Mini 1	Service comparison	Cloud models and trade-offs	Brief; working/source file or reproducible environment; output; validation notes; reflection.
Mini 2	Containerised workload	Compute, network and storage	Brief; working/source file or reproducible environment; output; validation notes; reflection.
Capstone	Cloud architecture	Security, resilience and operations	Brief; working/source file or reproducible environment; output; validation notes; reflection.

Standard project rubric

Dimension	Weight	What earns credit
Problem framing	15%	Clear objective, user/business need, assumptions and success criteria.
Technical/design accuracy	25%	Correct methods, appropriate tools and sound decisions.
Completeness	15%	Required features, assets, data or workflow are present.
Testing and validation	15%	Important normal, edge, quality or failure cases are checked.
Evidence and reproducibility	15%	Source/working files, setup, inputs, outputs and version information are available.
Documentation	5%	Concise structure, instructions and limitations.
Viva / ownership	10%	Learner can explain decisions, demonstrate work and respond to a change request.

Require least privilege, authorised labs, configuration evidence, logs, monitoring, rollback and documented troubleshooting.

NT-C-028 | Cloud Computing | Career Roles, Evidence & Gap Actions

Explainable top-three occupation matching

Rank	Occupation	Default weighted skills	Minimum evidence	Gap action
1	Cloud Support Trainee	Cloud concepts 50% [critical]; troubleshooting 50% [critical]	Lab portfolio	If a critical skill is below 60 or lab portfolio is missing, return Improve First or Needs Evidence with the linked module/project.
2	Junior Cloud Operations Associate	Monitoring 34% [critical]; access 33% [critical]; backup 33%	Operations runbook	If a critical skill is below 60 or operations runbook is missing, return Improve First or Needs Evidence with the linked module/project.
3	Infrastructure Support Trainee	Networking 34% [critical]; compute 33% [critical]; storage 33%	Architecture capstone	If a critical skill is below 60 or architecture capstone is missing, return Improve First or Needs Evidence with the linked module/project.

Course readiness	45% section mocks + 20% mini projects + 25% capstone + 10% viva.
Role readiness	45% role-relevant mastery + 25% projects/capstone + 20% aligned mocks + 10% verified resume evidence.
Evidence confidence	Show separately; Apply Now requires at least 70.
Resume extraction	Use only relevant skills, projects, education, experience and portfolio evidence; minimise personal data before external model processing.

A student may be Apply Now for one mapped occupation and Improve First for another. Return independent evidence and gaps for each role.

NT-C-028 | Cloud Computing | FAQs, Misconceptions & LLM Response Pack

Course-specific retrieval and response patterns

Approved FAQ	Answer
Who is Cloud Computing for?	Learners needing vendor-neutral cloud concepts before platform specialisation.
What should I know before starting?	Basic computer, networking and operating-system awareness.
How long is the approved learning journey?	10-14 weeks / 100-150 guided hours.
What will I be able to demonstrate?	Design and operate a secure, resilient cloud solution using core compute, storage, network and automation concepts.
Does completion guarantee a job?	No. Completion and role readiness are evidence-based guidance; employment, placement, salary and employer selection are not guaranteed.
What if I fail a section?	CareerTours identifies the failed skill tags and module, assigns targeted practice, and permits an equivalent retest after remediation.

Misconception	Approved correction
Attendance proves mastery	False - require tests and observable practical/project evidence.
One total score proves every role fit	False - calculate each occupation using its weighted skills and evidence.
The LLM can add missing details	False - retrieve approved records or state that evidence is insufficient.
Course completion guarantees employment	False - never claim guaranteed job, placement, salary or interview.
Cloud Computing knowledge never changes	False - version-sensitive tools, laws, taxes, security and platform procedures require review.

Retrieval metadata

```
course_code=NT-C-028; corpus_version=3.0;
record_types=course_profile,module,section,skill,assessment_sample,project,occupation_mapping,faq,misconception;
approval_status=approved; effective_date=2026-08-12
```